DERIVED KUNNETH ISOMORPHISM FOR BASE-FLAT QUASI-COHERENT SHEAVES OVER AN AFFINE BASE

THE STACKS PROJECT AUTHORS

Editorial note. No properness or Noetherian condition is added toX,Y,S. The source assumesX,Y quasi-compact with affine diagonal andF,G flat overS.

Editorial note. The ordinary field case invarieties.tex is not separately counted.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Quasi-coherent sheaves on schemes with affine diagonal, finite affine covers, cup products andCech resolutions, K-flat tensor products, Leray and flat cohomology base change. The localK-flat argument is retained.

Source reference. Original lemma 06Y2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8871–8871. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01FM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1875–1875. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 064K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7869–7869. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BED: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14346–14346. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 01FP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1878–1878. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original equation 07MB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9721–9721. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:perfect.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0FKV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19475–19475. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01XB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2512–2512. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0FLH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19497–19497. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01SG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2337–2337. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0FLM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19502–19502. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 089W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10511–10511. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01XL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2522–2522. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08EU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10685–10685. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0BEC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14345–14345. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0B68: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14061–14061. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

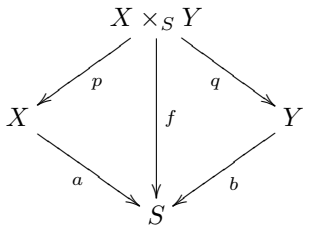
concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0FKU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19474–19474. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *In the situation above the map (1.0.3) is an isomorphism if S is affine, $\mathcal{F}$ and $\mathcal{G}$ are S-flat and quasi-coherent and X and Y are quasi-compact with affine diagonal.*

1. Künneth formula, II

For the case where the base is a field, please see Varieties, Section 0BEC. Consider a cartesian diagram of schemes



Let $K \in D(\mathcal{O}_X)$ and $M \in D(\mathcal{O}_Y)$. There is a canonical map

$$Ra_*K \otimes^{\mathbf{L}}_{\mathcal{O}_S} Rb_*M \longrightarrow Rf_*(Lp^*K \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_S Y}} Lq^*M) \tag{1.0.1}$$

Namely, we can use the maps $Ra_*K \to Ra_*Rp_*Lp^*K = Rf_*Lp^*K$ and $Rb_*M \to Rb_*Rq_*Lq^*M = Rf_*Lq^*M$ and then we can use the relative cup product (Cohomology, Remark 0B68).

Set $A = \Gamma(S, \mathcal{O}_S)$. There is a global Künneth map

$$R\Gamma(X, K) \otimes^{\mathbf{L}}_A R\Gamma(Y, M) \longrightarrow R\Gamma(X \times_S Y, Lp^*K \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_S Y}} Lq^*M) \tag{1.0.2}$$

in $D(A)$. This map is constructed using the pullback maps $R\Gamma(X, K) \to R\Gamma(X \times_S Y, Lp^*K)$ and $R\Gamma(Y, M) \to R\Gamma(X \times_S Y, Lq^*M)$ and the cup product constructed in Cohomology, Section 0FKU.

Let X, Y, S, a, b, p, q, f be as in the introduction to this section. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Let $\mathcal{G}$ be an $\mathcal{O}_Y$-module. Set $A = \Gamma(S, \mathcal{O}_S)$. Consider the map

$$R\Gamma(X, \mathcal{F}) \otimes^{\mathbf{L}}_A R\Gamma(Y, \mathcal{G}) \longrightarrow R\Gamma(X \times_S Y, p^*\mathcal{F} \otimes_{\mathcal{O}_{X\times_S Y}} q^*\mathcal{G}) \tag{1.0.3}$$

in $D(A)$. This map is constructed using the pullback maps $R\Gamma(X, \mathcal{F}) \to R\Gamma(X \times_S Y, p^*\mathcal{F})$ and $R\Gamma(Y, \mathcal{G}) \to R\Gamma(X \times_S Y, q^*\mathcal{G})$, the cup product constructed in Cohomology, Section 0FKU, and the canonical map $p^*\mathcal{F} \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_S Y}} q^*\mathcal{G} \to p^*\mathcal{F} \otimes_{\mathcal{O}_{X\times_S Y}} q^*\mathcal{G}$.

Lemma 1.1. *Let $f : X \to Y$ be a morphism of schemes with $Y = \mathrm{Spec}(A)$ affine. Let $\mathcal{U} : X = \bigcup_{i\in I} U_i$ be a finite affine open covering such that all the finite intersections $U_{i_0\ldots i_p} = U_{i_0} \cap \ldots \cap U_{i_p}$ are affine. Let $\mathcal{F}^\bullet$ be a bounded complex of $f^{-1}\mathcal{O}_Y$-modules. Assume for all $n \in \mathbf{Z}$ the sheaf $\mathcal{F}^n$ is a flat $f^{-1}\mathcal{O}_Y$-module and $\mathcal{F}^n$ has the structure of a quasi-coherent $\mathcal{O}_X$-module compatible with the given $p^{-1}\mathcal{O}_Y$-module structure (but the differentials in the complex $\mathcal{F}^\bullet$ need not be $\mathcal{O}_X$-linear). Then the complex $Tot(\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}^\bullet))$ is K-flat as a complex of A-modules.*

Proof. We may write

$$\mathcal{F}^\bullet = (\mathcal{F}^a \to \ldots \to \mathcal{F}^b)$$

Arguing by induction on $b - a$ and considering the distinguished triangle

$$\mathcal{F}^b[-b] \to (\mathcal{F}^a \to \ldots \to \mathcal{F}^b) \to (\mathcal{F}^a \to \ldots \to \mathcal{F}^{b-1}) \to \mathcal{F}^b[-b+1]$$

and using More on Algebra, Lemma 06Y2 we reduce to the case where $\mathcal{F}^\bullet$ consists of a single quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ placed in degree 0. In this case the Čech complex for $\mathcal{F}$ and $\mathcal{U}$ is homotopy equivalent to the alternating Čech complex, see Cohomology, Lemma 01FM. Since $U_{i_0 \ldots i_p}$ is always affine, we see that $\mathcal{F}(U_{i_0 \ldots i_p})$ is A-flat. Hence $\check{\mathcal{C}}^\bullet_{alt}(\mathcal{U}, \mathcal{F})$ is a bounded complex of flat A-modules and hence K-flat by More on Algebra, Lemma 064K. □

Proof. We strongly urge the reader to read the proof of Varieties, Lemma 0BED first. Choose finite affine open coverings $\mathcal{U} : X = \bigcup_{i \in I} U_i$ and $\mathcal{V} : Y = \bigcup_{j \in J} V_j$. This determines an affine open covering $\mathcal{W} : X \times_S Y = \bigcup_{(i,j) \in I \times J} U_i \times_S V_j$. Note that $\mathcal{W}$ is a refinement of $\mathrm{pr}_1^{-1}\mathcal{U}$ and of $\mathrm{pr}_2^{-1}\mathcal{V}$. Thus by the discussion in Cohomology, Section 01FP we obtain maps

$$\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}) \to \check{\mathcal{C}}^\bullet(\mathcal{W}, p^*\mathcal{F}) \quad \text{and} \quad \check{\mathcal{C}}^\bullet(\mathcal{V}, \mathcal{G}) \to \check{\mathcal{C}}^\bullet(\mathcal{W}, q^*\mathcal{G})$$

well defined up to homotopy and compatible with pullback maps on cohomology. In Cohomology, Equation (07MB) we have constructed a map of complexes

$$\mathrm{Tot}(\check{\mathcal{C}}^\bullet(\mathcal{W}, p^*\mathcal{F}) \otimes_A \check{\mathcal{C}}^\bullet(\mathcal{W}, q^*\mathcal{G})) \longrightarrow \check{\mathcal{C}}^\bullet(\mathcal{W}, p^*\mathcal{F} \otimes_{\mathcal{O}_{X \times_S Y}} q^*\mathcal{G})$$

which is compatible with the cup product on cohomology by Cohomology, Lemma 0FKV. Combining the above we obtain a map of complexes

$$\mathrm{Tot}(\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}) \otimes_A \check{\mathcal{C}}^\bullet(\mathcal{V}, \mathcal{G})) \to \check{\mathcal{C}}^\bullet(\mathcal{W}, p^*\mathcal{F} \otimes_{\mathcal{O}_{X \times_S Y}} q^*\mathcal{G}) \tag{1.1.1}$$

We claim this is the map in the statement of the lemma, i.e., the source and target of this arrow are the same as the source and target of (1.0.3). Namely, by Cohomology of Schemes, Lemma 01XB and Cohomology, Lemma 0FLH the canonical maps

$$\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}) \to R\Gamma(X, \mathcal{F}), \quad \check{\mathcal{C}}^\bullet(\mathcal{V}, \mathcal{G}) \to R\Gamma(Y, \mathcal{G})$$

and

$$\check{\mathcal{C}}^\bullet(\mathcal{W}, p^*\mathcal{F} \otimes_{\mathcal{O}_{X \times_S Y}} q^*\mathcal{G}) \to R\Gamma(X \times_S Y, p^*\mathcal{F} \otimes_{\mathcal{O}_{X \times_S Y}} q^*\mathcal{G})$$

are isomorphisms. On the other hand, the complex $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$ is K-flat by Lemma 1.1 and we conclude that $\mathrm{Tot}(\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}) \otimes_A \check{\mathcal{C}}^\bullet(\mathcal{V}, \mathcal{G}))$ represents the derived tensor product $R\Gamma(X, \mathcal{F}) \otimes^{\mathbf{L}}_A R\Gamma(Y, \mathcal{G})$ as claimed.

We still have to show that (1.1.1) is a quasi-isomorphism. We will do this using dimension shifting. Set $d(\mathcal{F}) = \max\{d \mid H^d(X, \mathcal{F}) \neq 0\}$. Assume $d(\mathcal{F}) > 0$. Set $U = \coprod_{i \in I} U_i$. This is an affine scheme as I is finite. Denote $j : U \to X$ the morphism which is the inclusion $U_i \to X$ on each U_i. Since the diagonal of X is affine, the morphism j is affine, see Morphisms, Lemma 01SG. It follows that $\mathcal{F}' = j_*j^*\mathcal{F}$ is S-flat, see Morphisms, Lemma 0FLM. It also follows that $d(\mathcal{F}') = 0$ by combining Cohomology of Schemes, Lemmas 089W and 01XB. For all $x \in X$ we have $\mathcal{F}_x \to \mathcal{F}'_x$ is the inclusion of a direct summand: if $x \in U_i$, then $\mathcal{F}' \to (U_i \to X)_*\mathcal{F}|_{U_i}$ gives a splitting. We conclude that $\mathcal{F} \to \mathcal{F}'$ is injective and $\mathcal{F}'' = \mathcal{F}'/\mathcal{F}$ is S-flat as well. The short exact sequence $0 \to \mathcal{F} \to \mathcal{F}' \to \mathcal{F}'' \to 0$ of flat quasi-coherent $\mathcal{O}_X$-modules produces a short exact sequence of complexes

$$0 \to \mathrm{Tot}(\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}) \otimes_A \check{\mathcal{C}}^\bullet(\mathcal{V}, \mathcal{G})) \to \mathrm{Tot}(\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}') \otimes_A \check{\mathcal{C}}^\bullet(\mathcal{V}, \mathcal{G})) \to \mathrm{Tot}(\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}'') \otimes_A \check{\mathcal{C}}^\bullet(\mathcal{V}, \mathcal{G})) \to 0$$

and a short exact sequence of complexes

$$0 \to \check{\mathcal{C}}^\bullet(\mathcal{W}, p^*\mathcal{F} \otimes_{\mathcal{O}_{X \times_S Y}} q^*\mathcal{G}) \to \check{\mathcal{C}}^\bullet(\mathcal{W}, p^*\mathcal{F}' \otimes_{\mathcal{O}_{X \times_S Y}} q^*\mathcal{G}) \to \check{\mathcal{C}}^\bullet(\mathcal{W}, p^*\mathcal{F}'' \otimes_{\mathcal{O}_{X \times_S Y}} q^*\mathcal{G}) \to 0$$

Moreover, the maps (1.1.1) between these are compatible with these short exact sequences. Hence it suffices to prove (1.1.1) is an isomorphism for $\mathcal{F}'$ and $\mathcal{F}''$. Finally, we have $d(\mathcal{F}'') < d(\mathcal{F})$. In this way we reduce to the case $d(\mathcal{F}) = 0$.

Arguing in the same fashion for $\mathcal{G}$ we find that we may assume that both $\mathcal{F}$ and $\mathcal{G}$ have nonzero cohomology only in degree 0. Observe that this means that $\Gamma(X, \mathcal{F})$ is quasi-isomorphic to the K-flat complex $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$ of A-modules sitting in degrees ≥ 0. It follows that $\Gamma(X, \mathcal{F})$ is a flat A-module (because we can compute higher Tor's against this module by tensoring with the Cech complex). Let $V \subset Y$ be an affine open. Consider the affine open covering $\mathcal{U}_V : X \times_S V = \bigcup_{i \in I} U_i \times_S V$. It is immediate that

$$\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}) \otimes_A \mathcal{G}(V) = \check{\mathcal{C}}^\bullet(\mathcal{U}_V, p^*\mathcal{F} \otimes_{\mathcal{O}_{X \times Y}} q^*\mathcal{G})$$

(equality of complexes). By the flatness of $\mathcal{G}(V)$ over A we see that $\Gamma(X, \mathcal{F}) \otimes_A \mathcal{G}(V) \to \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}) \otimes_A \mathcal{G}(V)$ is a quasi-isomorphism. Since the sheafification of $V \mapsto \check{\mathcal{C}}^\bullet(\mathcal{U}_V, p^*\mathcal{F} \otimes_{\mathcal{O}_{X \times Y}} q^*\mathcal{G})$ represents $Rq_*(p^*\mathcal{F} \otimes_{\mathcal{O}_{X \times Y}} q^*\mathcal{G})$ by Cohomology of Schemes, Lemma 01XL we conclude that

$$Rq_*(p^*\mathcal{F} \otimes_{\mathcal{O}_{X \times Y}} q^*\mathcal{G}) \cong \Gamma(X, \mathcal{F}) \otimes_A \mathcal{G}$$

on Y where the notation on the right hand side indicates the module

$$b^*\widetilde{\Gamma(X, \mathcal{F})} \otimes_{\mathcal{O}_Y} \mathcal{G}$$

Using the Leray spectral sequence for q we find

$$H^n(X \times_S Y, p^*\mathcal{F} \otimes_{\mathcal{O}_{X \times Y}} q^*\mathcal{G}) = H^n(Y, b^*\widetilde{\Gamma(X, \mathcal{F})} \otimes_{\mathcal{O}_Y} \mathcal{G})$$

Using Lemma 08EU for the morphism $b : Y \to S = \mathrm{Spec}(A)$ and using that $\Gamma(X, \mathcal{F})$ is A-flat we conclude that $H^n(X \times_S Y, p^*\mathcal{F} \otimes_{\mathcal{O}_{X \times Y}} q^*\mathcal{G})$ is zero for $n > 0$ and isomorphic to $H^0(X, \mathcal{F}) \otimes_A H^0(Y, \mathcal{G})$ for $n = 0$. Of course, here we also use that $\mathcal{G}$ only has cohomology in degree 0. This finishes the proof (except that we should check that the isomorphism is indeed given by cup product in degree 0; we omit the verification). $\square$